INCH-POUND
MIL-DTL-32744
25 May 2023

DETAIL SPECIFICATION

BIS(2,2-DINITROPROPYL)ACETAL AND BIS(2,2-DINITROPROPYL)FORMAL, MIXTURE OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations, and tests for a 50:50 mixture of bis(2,2-dinitropropyl)acetal (BDNPA) and bis(2,2-dinitropropyl)formal (BDNPF) and containing diphenylamine stabilizer, referred to as BDNPA/F.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract..

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-98 - Diphenylamine

Comments, suggestions, or questions on this document should be addressed to: U.S. Army DEVCOM Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000, or emailed to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-286	-	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Non-Government standards and publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E300	-	Standard Practice for Sampling Industrial Chemicals
ASTM E617	-	Standard Specification for Laboratory Weights and Precision Mass Standards
ASTM D1193	-	Standard Specification for Reagent Water

(Copies of these documents are available from https://www.astm.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.3), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.2 Physical and chemical properties. The physical and chemical requirements of the mixture of BDNPA and BDNPF (BDNPA/F) shall be as specified in Table I.

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TABLE I. Physical and chemical requirements for BDNPA/F.

Property	Requirement
Acid number (mg KOH/g) maximum (max.)	0.20
Moisture content, weight percent (wt. %) max.	0.05
BDNPA content (wt. %)	45.0 – 55.0
BDNPF content (wt. %)	45.0 – 55.0
Diphenylamine content (wt. %) 1/	0.10 – 0.22
Extraction solvent content (wt. %) max.	0.04
Acid number after accelerated aging (mg KOH/g) max.	0.50
Notes: 1/ Diphenylamine shall conform to the requirements of MIL-DTL-98.	

3.3 Workmanship. BDNPA/F shall be uniform, free from visible impurities, and foreign material such as dirt, wooden particles or any other foreign matter.

4. VERIFICATION

TABLE II. Requirements/verification cross reference matrix.

Method of verification 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test						Classes of verification A – First article B – Conformance		
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article	X		X		X		4.2
3.1.2	Conformance	X		X			X	4.3
3.2	Acid number	X				X	X	4.4.1
3.2	Moisture content	X				X	X	4.4.2
3.2	BDNPA content	X				X	X	4.4.3
3.2	BDNPF content	X				X	X	4.4.3
3.2	Diphenylamine content	X				X	X	4.4.4
3.2	Extraction solvent content	X				X	X	4.4.5
3.2	Acid number after accelerated aging	X				X	X	4.4.6
3.3	Workmanship			X		X	X	4.5

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4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see paragraph 4.2).
b. Conformance inspection (see paragraph 4.3).

4.2 First article inspection. When specified (see 6.2 and 6.8), a sample shall be subjected to first article verification in accordance with 4.3.3. A sample of not less than 600 mL shall be submitted for examination. Samples shall be produced by the contractor using the same process procedures, material supply sources, production equipment, and production line as will be used in regular production.

4.2.1 Rejection. If any sample fails to comply with any of the applicable requirements the first article quantity shall be rejected.

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots and batches shall comply with the MIL-STD-1916, paragraph "Formation and identification of lots or batches." Each batch shall consist of that quantity of BDNPA/F that has been subjected to the same chemical and physical mixing process intended to make the final product homogeneous. A lot shall consist of one or more acceptable batches of BDNPA/F produced by one manufacturer in accordance with one specification under one continuous set of operating conditions. When multiple acceptable batches are combined into a lot the individual batches shall be blended together to ensure homogeneity.

4.3.2 BDNPA/F sampling. Samples of not less than 300 mL shall be collected from the beginning, middle, and end of every batch in every lot in accordance with ASTM E300. A composite sample shall not be permitted. If any sample fails to comply with any applicable requirements the batch shall be rejected.

4.3.3 First article and classification of characteristics.

Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Acid number	4.3.2	3.2	4.4.1
102	Moisture content	4.3.2	3.2	4.4.2
103	BDNPA content	4.3.2	3.2	4.4.3
104	BDNPF content	4.3.2	3.2	4.4.3
105	Diphenylamine content	4.3.2	3.2	4.4.4
106	Extraction solvent content	4.3.2	3.2	4.4.5
107	Acid number after accelerated aging	4.3.2	3.2	4.4.6 and 4.4.1
Minor				
201	Workmanship	4.3.2	3.3	4.5

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4.4 Test methods and procedures. Unless otherwise specified, all chemicals and reagents shall be reagent grade or better, and water shall be distilled, deionized, or equivalent, conforming to ASTM D1193 Type I or Type II. Analytical balance weights shall be conforming to ASTM E617.

4.4.1 Acid number. Determine acid number in accordance with the following procedure.

4.4.1.1 Apparatus and reagents.

a. Automatic recording titrator with a combination electrode or a pH measuring electrode with glass-calomel reference and corresponding micro-buret.
b. Several 250 mL Erlenmeyer flasks and 150 mL beakers
c. Chloroform (stabilized)
d. Methanol
e. Tetrabutyl ammonium hydroxide
f. Isopropyl alcohol

4.4.1.2 Standard preparation.

a. Prepare 0.05N and 0.1N solutions of tetrabutyl ammonium hydroxide in isopropyl alcohol, standardized to four decimal places against potassium acid phthalate.
b. Standardize operation and preamp of the titrator in accordance with the manufacturer's instruction.
c. Buffer the electrodes with pH 10 buffer.
d. Titrate at maximum rate while purging air from buret tip.
e. Condition the electrodes by soaking in water between titrations.
f. Do not titrate the sample past an apparent pH of 11.0.
g. The potentiometric end point is best determined by a set of concentric circles.

4.4.1.3 Procedure.

a. Weigh 10 grams of BDNPA/F to the nearest 0.005 gram into a 150 mL beaker.
b. Add 80 mL of chloroform and 20 mL of methanol to the beaker. Mix with a magnetic stir bar until the sample is dissolved.
c. Rinse the buret tip and electrode with methanol and then water to keep the pH electrode hydrated. If the titration curve becomes erratic, soak the electrodes in a pH 4 buffer for several minutes before beginning another titration.
d. Place the sample beaker into position and adjust the elevation of the beaker platform so that the buret tip and electrode are immersed in the solution.
e. Adjust the stirrer for proper speed.
f. Titrate the solution with 0.05N tetrabutyl ammonium hydroxide (TBAH) solution approximately 0.2 mL beyond an inflection. NOTE: If 1 mL titration is exceeded before pH 11 or an inflection is reached, change to a 5 g sample and 0.1N titrant then re-start procedure from 4.4.1.3.a.
g. Determine the volume of the titration to the nearest 0.01 mL directly from the chart.

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h. Repeat steps a through g for each additional sample. The samples should be run in triplicate, where solvent is added immediately prior to each individual titration to produce consistent results.

i. Titrate a blank to the potentiometric end point of the sample titration.

4.4.1.4 Calculation. Calculate acid number as milligrams of KOH per gram (mg KOH/g) per the following equation:

$$\text{Acid number} = \frac{(V_1 - V_2) \times N \times 56.1}{W}$$

Where: V_1 = Volume required to titrate the sample solution, mL
V_2 = Volume required to titrated the blank solution, mL
N = Normality of TBAH solution (meq/mL)
W = Weight of sample, g

4.4.2 Moisture. Determine moisture content in accordance with Method 101.5 Moisture (Karl Fischer Method), of MIL-STD-286. Sample size shall be between 8 and 12 grams and the carrier solution shall be anhydrous methanol.

4.4.3 BDNPA and BDNPF analysis.

4.4.3.1 Apparatus for synthesis of high purity standards. Set up a 2-inch Arthur F. Smith high vacuum wiped-film still as follows.

a. Set up the regular vapor cold trap and diffusion pump supplied with the unit as directed.

b. Attach the manometer between the still and the vapor trap.

c. Attach a vacuum pump protected by two cold traps in series to the diffusion pump.

d. Secure a 500 mL round bottom flask receiver constructed with a bottom and overhead outlet to the residue outlet at the bottom of the still. Connect the bottom outlet of this flask to the top outlet of a similar flask just below a ¼-inch latex vacuum hose equipped with a pinch clamp.

e. By means of rubber tubing, use the top outlet of the residue receiver to connect a pressure equalizing line to the neck of the lower feed container.

f. Use a sigmamotor pump to transfer the standard from the bottom outlet of the feed container through a glass condenser heat exchanger, to the feed inlet of the wiped-film still through ¼-inch latex vacuum tubing.

g. Cool both the residue received and the feed container by circulating cold water in and out of the cooling pans supporting these flasks.

h. Attach a suitable distillate receiver to the distillate outlet of the still.

i. Circulate the fluid through the condenser finger in the still and heat exchanger by means of an immersion pump.

j. Wrap an electric heating mantle around the still and measure the skin temperature between the still and the mantle by means of a thermocouple temperature indicator.

k. Check all connections for leaks prior to charging the feed container with the standard.

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l. Charge the cold traps with dry ice and activate the vacuum pump.

m. Turn on the diffusion pump when the pressure is 1 to 2 mm Hg.

n. Adjust the sigmamotor pump to 1 to 1.5 liters per hour and activate the still's wiping blades.

o. Adjust the still heating mantle to 120°C to 125°C skin temperature and the condenser and the heat exchanger fluid to 40°C to 50°C.

p. The first pass through the still is usually sufficient to degas the standard.

q. When the feed container is empty, release the pinch clamp and let the degassed material from the receiver enter the feed container.

r. After the standard is completely degassed and distilling properly, the pinch clamp can be removed; but if left in, the number of passes through the still to complete the distillation can be determined.

4.4.3.2 Synthesis of high purity standards. Fresh standards shall be synthesized not less than monthly.

4.4.3.2.1 High purity BDNPA synthesis.

a. Combine 450.3 grams (3.0 moles) of 2,2-dinitropropanol (DNPOH) and 105.5 grams (2.4 moles) of acetaldehyde in 1470 grams of ethylene chloride in a three-necked flask equipped with cooling bath, thermometer, and dropping funnel.

b. Stir the solution continuously, keeping the temperature between -9°C and -11°C.

c. Add 213 grams (1.5 moles) of boron trifluoride etherate over a 15 minute period as follows:

i. Add 15 to 30 grams of boron trifluoride etherate.

ii. Cool until the exothermic reaction is under control.

iii. Add the remainder of boron trifluoride etherate.

d. Stir the mixture for 15 minutes and then quench with 930 grams of water.

e. Agitate for ten minutes then allow the two phase mixture to settle.

f. Separate the aqueous layer and discard. Wash the organic layer with sufficient 4% sodium hydroxide solution to raise the pH to 11.0.

g. Control the pH at 11.0 by intermittently adding sodium hydroxide over 20 to 30 minutes of stirring.

h. Separate the layers: discard the aqueous layer, and wash the organic layer again with an equal volume of 5% sodium hydroxide solution at 10°C to 15°C for 30 minutes.

i. Separate the layers: discard the aqueous layer, and wash the organic layer with one liter portions of water until neutral.

j. Remove the solvent under vacuum at 45°C and 0.1 mmHg.

k. Dissolve the crude BDNPA in 0.84 grams of isopropyl ether per 1.0 gram of BDNPA and crystallize at -30°C for 2 hours.

l. Filter the crystals by suction and dry at 45°C and 0.1 mm Hg for 2 hours.

m. Charge the crystallized BDNPA into a 2 inch Arthur F. Smith high vacuum still and distill twice at 125°C and 10 µm Hg.

4.4.3.2.2 High purity BDNPF synthesis.

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a. Combine 473.0 grams (3.15 moles) of 2,2-dinitropropanol (DNPOH) and 45 grams (0.5 moles) of trioxane in 1470 grams of ethylene chloride in a three-necked flask equipped with cooling bath, thermometer, and dropping funnel.

b. Add 225 grams (2.25 moles) of 98% sulfuric acid over a 15 minute period as the solution continuously stirs.

c. Stir the mixture for 45 minutes at 25°C.

d. Allow the two phase mixture to settle.

e. Separate the sulfuric acid layer and discard. Wash the organic layer twice with one half of its volume of 5% sodium hydroxide. During the two washings, dwell for 30 minutes at 15°C and 30°C, respectively.

f. Separate and discard the aqueous layer after each washing.

g. Wash the organic layer again with one liter portions of water until neutral.

h. Remove the solvent under vacuum at 45°C and 0.1 mmHg for 2 hours.

i. Dissolve the crude BDNPF in 1.4 grams of denatured alcohol (a mixture of 100 parts by volume No. 1 denatured ethanol, 10 parts isopropanol, and 1 part methyl isobutyl ketone) per 1.0 gram of BDNPF and crystallize at -30°C for 2 hours

j. Filter the crystals by suction and dry at 45°C and 0.1 mm Hg for 2 hours.

k. Charge the crystallized BDNPF into a 2 inch Arthur F. Smith high vacuum still and distill twice at 125°C and 10 µm Hg.

4.4.3.2.3 Apparatus and reagents for analysis.

a. An HPLC with the following equipment suggested (any comparable equipment may be substituted:

i. Waters 2695 Separation module with column heater

ii. Waters 2996 Photo Diode Array (PDA) detector with 10 mm cell

iii. Kanto Mightysil RP-18GP 250x4.6 mm 3um column

b. Solvents for HPLC use, i.e. acetonitrile, methanol, and water, shall be HPLC-grade.

4.4.3.2.4 Preparation of standards and calibration procedure.

a. Transfer 0.125 ± 0.010 g each of BDNPA and BDNPF standards obtained in 4.4.3.1 into a 50 mL volumetric flask. Record the weights to the nearest 0.0001 g.

b. Dilute with 30-40 mL HPLC-grade acetonitrile and swirl well to mix. Do NOT make up to volume.

c. Prepare a 0.5 mg/mL stock solution of diphenylamine (DPA) by the following steps.

i. To a 50 mL volumetric flask add 0.0250 ± 0.0030 g reagent-grade DPA, recording the weight to the nearest 0.0001 g.

ii. Add about 40 mL HPLC-grade acetonitrile, swirl to dissolve, and then make up to volume with additional acetonitrile and shake well to mix.

d. Use a volumetric pipette to transfer 1.00 mL of the DPA stock to the BDNPA/F standard.

e. Make up the composite standard up to volume with HPLC-grade acetonitrile and shake well to mix.

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f. Transfer solution to HPLC vial and analyze along with the samples (See 4.4.3.2.5.a-c) per the parameters suggested in Table IV.

g. Extract and integrate chromatograms at 237 nm.

h. From injections of the standard, identify the BDNPA and BDNPF peaks based on peak height and retention time. BDNPA and BDNPF shall be identified as the two tallest peaks (expected height 0.5 to 1.2 absorbance units) in the chromatogram; BDNPA elutes at 13.4 min and BDNPF at 12.5 min when run as per Table IV. Note: retention times may be different for a particular set-up used.

i. For each analyte, note the average retention time and its range over the three injections.

j. Obtain inverse response factors (IRF) for each of the analytes (i.e. BDNPA and BDNPF) based on peak area as follows; use scientific notation and calculate to at least five (5) significant figures:

$$IRF = \frac{W}{A}$$

Where: IRF = inverse response factor
W = weight of analyte in 50 mL, g
A = analyte peak area

k. For each analyte, calculate a mean IRF over the three injections.

TABLE IV. HPLC instrument parameter guide for BDNPA/F analysis.

Waters 2996 photo diode array parameters				
Wavelength Range	200-400 nm			
Extraction / integration wavelength	237 nm for BDNPA and BDNPF, 283 nm for DPA			
Resolution	1.2 nm			
Waters 2695 separation module parameters				
Column Temperature	40ºC			
Eluent Flow Rate	0.9 mL/min			
Run Time	18 min			
Pre-injection equilibration	5 min at initial conditions			
Injection Volume	10 µL			
Number of Injections	3 per standard, 2 per sample			
Eluent Composition, % volume				
Time, minutes	Water	Methanol	Acetonitrile	Gradient description
0	60	10	30	Initial condition, no hold
10	20	50	30	Linear change from previous, then hold 5 min
15	60	10	30	Step change at 15 min, hold 3 min to end

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Note 1: Solvents shall be HPLC-grade. Water shall be HPLC-grade or ASTM D1193 Type I or Type II. Note 2: HPLC parameters may be adjusted to maintain adequate resolution and shape of the component peaks.

4.4.3.2.5 Preparation of samples and quantitation procedure.

a. Transfer 0.25 ± 0.01 g of the composite BDNPA/F sample into a 50 mL volumetric flask. Record the sample weight to the nearest 0.0001 g. Prepare a duplicate the same way.

b. Dilute both samples to volume with HPLC grade acetonitrile and shake well to mix.

c. Transfer solutions to HPLC vials and analyze along with the standard (See 4.4.3.2.4.e-f) per the parameters suggested in Table IV.

d. Extract and integrate chromatograms at 237 nm.

e. Identify the BDNPA and BDNPF peaks based on peak size and retention time: the analytes shall be the largest peaks with retention times in the same time range as the standards (See 4.4.3.2.4.k).

f. For each sample injection, calculate the content of each analyte from its peak area and its IRF as follows:

$$C = \frac{100 \times A \times IRF}{S}$$

Where: C = content of BDNPA or BDNPF, weight percent
A = peak area of BDNPA or BDNPF
IRF= inverse response factor for BDNPA or BDNPF (see 4.4.3.2.4.k)
S = sample weight, g (see “a”)

g. Calculate and report the mean content of each, BDNPA and BDNPF, by averaging the results from 4.4.3.2.5.f across the four injections (two injection each for two samples).

4.4.4 Diphenylamine content analysis. Diphenylamine content is obtained from the HPLC data previously collected during BDNPA/F purity analysis (see 4.4.3) in accordance with the following procedures.

a. Re-examine the HPLC data collected for BDNPA/F purity analysis in 4.4.3.

b. Extract and integrate chromatograms at 283 nm.

c. Examine all injections and identify the DPA peak in each based on its retention time and its ultraviolet absorption spectrum. DPA elutes soon after BDNPA; its retention time is approximately 13.9 min when run under the conditions in Table V, and its UV absorption spectrum above 220 nm has a single local maximum at 283 ± 4 nm.

d. For each sample injection, calculate the weight percentage of DPA as follows:

$$\text{Diphenylamine} = \frac{2 \times A \times D}{R \times S}$$

Where: A = peak area of DPA in sample injection

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D = weight of DPA used in stock, g (see 4.4.3.2.4.c)
R = peak area of DPA in standard, averaged over three injections
S = sample weight, g (see 4.4.3.2.5.a)

e. Calculate and report the mean weight percentage of diphenylamine by averaging the results from "d" across the four sample injections (two injection each for two samples).

4.4.4.1 Diphenylamine. Verify by examination of objective evidence that diphenylamine is in accordance with MIL-DTL-98.

4.4.5 Extraction solvent content analysis. The following procedure is used to determine the amount of residual organic solvent(s) in the BDNPA/F sample using gas chromatography – mass spectrometry (GC-MS) analysis of its solution in n-butyl acetate.

4.4.5.1 Reagents and chemicals.

a. n-butyl acetate, reagent grade
b. other solvents (as needed, see 4.4.5.5 and 4.4.5.6.d), reagent grade

4.4.5.2 Supplies and equipment.

a. Gas Chromatography - Mass Spectrometry (GC-MS) instrument, e.g. ThermoFisher Focus GC, DSQ II, AS3000 autosampler or equivalent, with computer system and NIST electron ionization spectral library.
b. Gas chromatographer (GC) capillary column, e.g. Agilent PoraBond Q 25 m x 0.25 mm x 0.25 um or equivalent
c. Compressed helium: 99.999% pure or better
d. Analytical balance: 200 g capacity, 0.001 g minimum accuracy
e. Volumetric flasks, glass: 1 x 5 mL, 6 x 10 mL, 1 x 100 mL
f. Volumetric pipettes, glass: one 1 mL "To deliver"; one 2 mL graduated in 0.1 mL increments; both Class A (see 6.7)
g. Adjustable volume pipettes: 1 – 5 mL; 5 – 50-microliter (µL); 50 – 200-µL; with disposable plastic tips

4.4.5.3 Instrument conditions. The following conditions are recommended, but may be adjusted slightly to accommodate a particular instrumental set-up. Once optimized, however, the same conditions shall be used for analyses of all samples and standards (see 4.4.5.8 and 4.4.5.9).

a. Injection port: 200˚C, 20-fold split (a split liner with a glass wool plug is recommended), injection volume – 2 µL (Use a 5-10 s needle dwell in injection port prior to injection).
b. Oven program: 40˚C hold for 5 min, ramp at 20˚C/min to 240˚C, hold for 5 min.
c. Transfer line to Mass Spectrometer (MS) (see 4.4.4.2.a) at 300˚C.
d. MS source: 200˚C, 70 eV in positive electron impact mode, collect m/z range 10 – 510 at 5 Hz.

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4.4.5.4 Initial assessment.

a. Dissolve 1.5 ± 0.1 g (approximately 1.1 mL) of sample in a 5 mL volumetric flask in n-butyl acetate. Ensure that the sample is fully dissolved.
b. Transfer sample solution to a GC vial.
c. Fill a separate vial with n-butyl acetate (solvent blank).
d. Analyze both sample and blank using conditions in section 4.4.5.3 and identify the components per section 4.4.5.5.

4.4.5.5 Identify components. Perform this analysis to identify solvent components as part of first article test (see 4.2), or following a change in process or supplier.

a. Review the collected total ion current chromatograms for sample and blank. Since butyl acetate peaks dominate the chromatograms, zoom to a section that excludes the largest of such peaks and re-scale the intensity.
b. Identify peaks belonging exclusively to the sample, i.e. those present in the sample but absent in the blank. If no peaks are detected, extract and examine a chromatogram at m/z range 30-100.
c. For each peak found, extract a baseline-subtracted mass spectrum, and identify the compound by comparison to the NIST EI library.
d. For each compound identified record the retention time of its chromatographic peak, and the m/z value of the most prominent peak in its mass spectrum, one that yields a chromatogram with the highest signal-to-noise ratio for the given compound.

NOTE: If no peaks were detected in the sample chromatogram, the sample shall be deemed free of residual solvent. Per 4.4.5.9.f report the solvent content. Repeat 4.4.5.5 for the next batch inspected.

4.4.5.6 Preparation of calibration standards. Each solvent identified in section 4.4.5.5 shall be used to create a composite stock solution as follows.

a. To a 100 mL volumetric flask, add 90 – 95 mL n-butyl acetate and cap the flask. Place the flask on a balance and zero it.
b. Add 0.100 ± 0.020 g of the first solvent. Record the weight and name of solvent added, and then re-zero the balance.
c. Repeat step “b” for each of the remaining solvents. Finally, make up the flask to volume with n-butyl acetate and invert five times to mix. Calculate the exact concentration of each analyte to the nearest 0.00001 g/mL.
d. Select a compound to use as Internal Standard (IS). The choice shall be a solvent NOT found in the sample, e.g. acetonitrile, methanol. Moreover, it shall be a single compound (e.g. not a mixture of hexane isomers) with a normal boiling point in the 36-90˚C range, which is fully miscible with n-butyl acetate. Additionally, when viewing chromatograms extracted using one of its prominent mass-spectral signals, the IS shall be sufficiently resolved from all chromatographic peaks belonging to sample analytes or n-butyl acetate. This may require analyzing the sample solutions in butyl acetate spiked with candidate compounds using the conditions in 4.4.5.3.

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e. Once an appropriate IS is chosen, record the retention time of its chromatographic peak, and the m/z value of the most prominent signal in its baseline-subtracted mass spectrum, one that yields a chromatogram with the highest signal-to-noise ratio for the IS peak.

f. Prepare an IS stock solution. To a 10 mL volumetric flask add 9 mL n-butyl acetate. Add approximately 0.02 – 0.03 g of the neat IS selected in "d" (the exact weight used need not be known as long as the same IS solution is used to prepare all samples and working standards). Fill up to volume with n-butyl acetate and invert five times to mix.

g. Prepare working standards. Use a graduated pipette to transfer 0.50 mL, 1.00 mL, and 1.50 mL aliquots of the stock prepared in 4.4.5.6.c to three separate 10 mL volumetric flasks. Use a volumetric pipette to add 1.00 mL of the IS stock prepared in "e" to each of the flasks. Calculate the exact amount of each analyte in each solution to the nearest 0.00001 g.

h. Transfer the standard solutions to GC vials. Fill another vial with n-butyl acetate to serve as a blank.

4.4.5.7 Sample preparation.

a. To a 10 mL volumetric flask, add 3.00 ± 0.10 g of sample (about 2.2 mL) and record the sample weight to the nearest 0.01 g. Add 5 mL n-butyl acetate and 1.00 mL of the IS stock (4.4.4.7 e). Cap and shake to completely dissolve, then make up to volume with n-butyl acetate. Cap and shake to mix.

b. Repeat step 4.4.5.7.a to prepare a second sample.

c. Transfer both solutions to GC vials.

4.4.5.8 Analysis.

a. Set up the GC-MS as per 4.4.5.3.

b. Make three (3) injections of each of the standards, followed by one (1) of the blank (4.4.5.6.h), and one (1) of each sample (4.4.5.7.c).

4.4.5.9 Calibration and quantitation. The following tasks are best carried out using computer software included with the GC-MS system and a spreadsheet.

a. For each of the analytes and the IS, extract chromatograms using m/z signals previously selected (4.4.5.5.c, 4.4.5.6.e). Integrate peak areas for each analyte and IS. Calculate the ratio of analyte and IS peak area. This shall be done for all injections (4.4.5.8.b). Check the blank injection and make sure there is no carryover of analytes from the standards.

b. Using the data from the standards, for each analyte, perform a least-squares linear regression relating the quotients found in "a" (abscissa) to the corresponding amounts of analyte present (ordinate) (see 4.4.5.6.g). This will yield a number of calibration lines, of the form $y=mx+b$, equal to the number of analytes present. Ensure that the correlation coefficient R^2 for each of the lines is greater than 0.99.

c. Use the peak area ratio data from 4.4.5.9.a for the samples and the calibration lines from 4.4.5.9.b to determine the amount of each analyte in the sample solutions to the nearest 0.00001 g.

d. Determine the percent weight of each solvent in the sample to the nearest 0.001% as follows:

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$$\text{Percent solvent} = 100 \times \frac{A}{W}$$

Where:
A = amount of a given solvent found in 4.4.5.9.c, grams
W = sample weight from 4.4.5.7.a, grams

e. For each analyte, average the results from the duplicate sample preparations.
f. Sum the results from 4.4.5.9.e and report the sum as Extraction solvent.

4.4.6 Acid number after accelerated aging. The test for determining acid number shall be as specified in section 4.4.1, but on a sample aged as follows:

a. Weigh 100 grams of the BDNPA/F mixture into a 250 mL Erlenmeyer flask.
b. Cover flask with aluminum foil.
c. Heat flask in an oil bath or safety oven for a minimum of 168 hours, while maintaining the temperature at 104 ± 0.5°C.

4.5 Workmanship. Place approximately 100 mL of BDNPA/F into a 150 mL glass beaker over a clean surface with back-lighting. The sample shall be examined by personnel with normal vision or normal corrected vision. The sample shall be uniform and free from visible impurities, and foreign material such as dirt, wooden particles or any other foreign matter. Failure to meet the workmanship requirement during inspection or while executing production processes shall result in batch rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point' packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. BDNPA/F is intended for use as an ingredient in propellant for rocket motors and as a plasticizer in explosive formulations.

6.2 Acquisition requirements. Acquisition documents should specify the following:

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a. Title, number, and date of this specification.
b. If required, the specific issue of individual documents referenced (see 2.1 and 2.2).
c. Provisions for submission/resubmission of first article samples.
d. Quantity required and delivery schedules.
e. Levels of preservation and packing (see 5.1).

6.3 Consideration of data requirements. When this specification is used in a procurement contract which incorporates DD Form 1423, contract data requirements list, acceptance and description sheets (for explosives) are required for each lot. These sheets should be prepared and submitted by the contractor in accordance with MIL-STD-1171 or otherwise as directed by the contract.

6.4 Technical agency. DEVCOM Armaments Center is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: DEVCOM Armaments Center, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

6.5 Approval of equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: DEVCOM Armaments Center, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.6 Submission of alternative conformance inspection provisions. Unless otherwise specified in the contract, proposed alternative conformance inspection provisions should be submitted by the contractor for evaluation by the Technical Agency responsible for preparation of this specification. (See 6.4).

6.7 Class A glassware. Class A glassware should meet the requirements of ASTM E694 “Standard Specification for Laboratory Glass Volumetric Apparatus” and ASTM E542 “Standard Practice for Calibration of Laboratory Volumetric Apparatus.”

6.8 Conditions requiring first article. First article inspection should be required whenever a change occurs in the manufacturing process, material used, drawing, specification, or source of supply that might significantly affect product quality as determined by the Government, or whenever there is a lapse in BDNPA/F production for a period in excess of six months.

6.9 Incorporated changes. ECP Y21Q2008 provides documentation.

6.10 Subject term (key word) listing.

Energetic material
Propellant ingredient
Plasticizer

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CONCLUDING MATERIAL

Custodians:
Army-AR
Navy-AS

Preparing activity
Army-AR
(Project 6810-2021-014)

Review activities:
Army-EA, MD, MI
Navy-OS
DLA-GS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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